

Phase 5 - README

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PHASE 5 — Evidence Intelligence & RAG

Real Estate AI Decision Intelligence Platform

Phase 4 (Predictive & Advanced Analytics) ke baad ka agla step — Master A-Z Prompt ki 10-phase architecture me Phase 5. Yeh phase Phase 3 ke code-structure convention ko exactly follow karta hai (live PostgreSQL via `db/connection.py`, `src/<topic>/*.py` modules, do orchestrators — Markdown report aur numbered CSV export — plus chart PNGs).

Kya banaya gaya

Module	Kaam	Master Prompt Section
<code>evidence_registry.py</code>	Evidence Registry (120 documents, <code>ai.documents</code> se)	5.3
<code>chunking.py</code>	Sentence-aware chunking, document relationships	5.4
<code>lexical_retrieval.py</code>	BM25 (keyword-based)	5.5
<code>semantic_retrieval.py</code>	LSA (offline semantic signal — transformer embedding NAHI)	5.5
<code>hybrid_retrieval.py</code>	BM25 + LSA combine, metadata filtering	5.5
<code>reranking.py</code>	Source quality + recency + query alignment	5.6
<code>grounding.py</code>	Citation ID validation, claim validation, coverage	5.7
<code>prompt_injection.py</code>	11-pattern injection/authority-spoofing detector	5.8
<code>observability.py</code>	Query/latency/zero-result-rate logging	5.9
<code>rag_evaluation.py</code>	6 question categories	5.10

Poori detail: `docs/phase5_report.md` (phase completion report — re-check me mile 3 bugs bhi documented hain), `docs/limitations.md`.

Kaise chalayein

Live database ke saath (production path — jaisa Phase 3 aapke local environment

me chala)

```
cd phase5
pip install -r requirements-phase5.txt --break-system-packages
export DB_USER=...
export DB_PASSWORD=...

python3 src/evidence/generate_evidence_report.py # docs/evidence_report.md
python3 src/evidence/export_results_csv.py # docs/evidence_results_csv/*.csv
python3 src/evidence/generate_charts.py # docs/evidence_charts/*.png
```

Local re-check (bina database ke — jo maine yahan use kiya)

```
python3 src/evidence/generate_evidence_report.py --source csv
python3 src/evidence/export_results_csv.py --source csv
python3 src/evidence/generate_charts.py --source csv
python3 src/evidence/test_phase5.py # 27 automated tests
```

--source csv data/processed/documents.csv se padhta hai — yeh wahi rows hain jo ai.documents table me load hui thi. Isliye dono paths (db aur csv) same evidence registry content produce karte hain — sirf transport alag hai.

Re-check me kya mila (important)

Maine sirf modules likh kar chhoda nahi — poora pipeline real 120-document data par chalaya aur test kiya. Isme **3 real bugs mile aur fix kiye**:

1. Grounding module citation marker ([ev_1_0]) ke andar ke numbers (1, 0) ko galti se “claim” samajh raha tha — fix kiya.
2. Report generator ka demo text banane wala code . par naive split kar raha tha, jo decimal numbers (jaise 59.9) ko bhi tod raha tha — fix kiya.
3. Citation ko sentence ke period ke baad append karne se ek alag “orphan sentence” ban raha tha jisme citation tha par claim nahi — fix kiya.

Poori detail docs/phase5_report.md Section 8 me hai. Fix ke baad **27/27 tests pass**.

Folder structure

```
phase5/
├─ data/processed/      # documents.csv (120 rows) + supporting tables
├─ db/connection.py     # Phase 1/3 se unchanged – env-var based Postgres pool
├─ src/evidence/        # 10 core modules + 3 orchestrator scripts + tests
├─ docs/               # report, limitations, 8 CSVs, 4 charts
├─ requirements-phase5.txt
└─ README_PHASE5.md
```

Next phase dependency

Phases 6/7/8 (Financial, Risk, Geo) is phase par strictly depend nahi karte, par optionally is phase ki grounding API use kar sakte hain evidence cite karne ke liye. Phase 9 ka Evidence Agent aur Evidence Validator seedha hybrid_retrieval.py, reranking.py, grounding.py, prompt_injection.py import karenge.